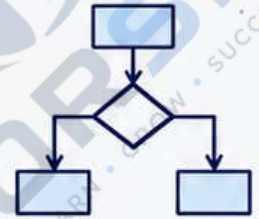


1010101010
0101010101
1010101010
0101010101



INSTRUCTIONS



1. Read the Problem Statement

Understand the objective and expected output before writing the program.



2. Write and Run the Program

Implement the solution, compile and run the program to check the output.



3. Test with Different Inputs

Use multiple test cases to verify the correctness of the program.



4. Add Proper Comments

Include comments to explain the logic and important steps.

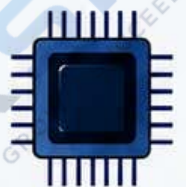


5. Maintain the Record

Write the program, output and observations neatly in your practical file.

Code • Practice • Learn • Improve

HAPPY PRACTICALS !



10110
01001
10101



Practical No-01

Aim: Write a program in Java to Swap Two Numbers using third Variables.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 14/07/2026

Source Code:

```
public class SwapNum
{
    public static void main(String[] args)
    {
        int a = 10, b = 20, c;

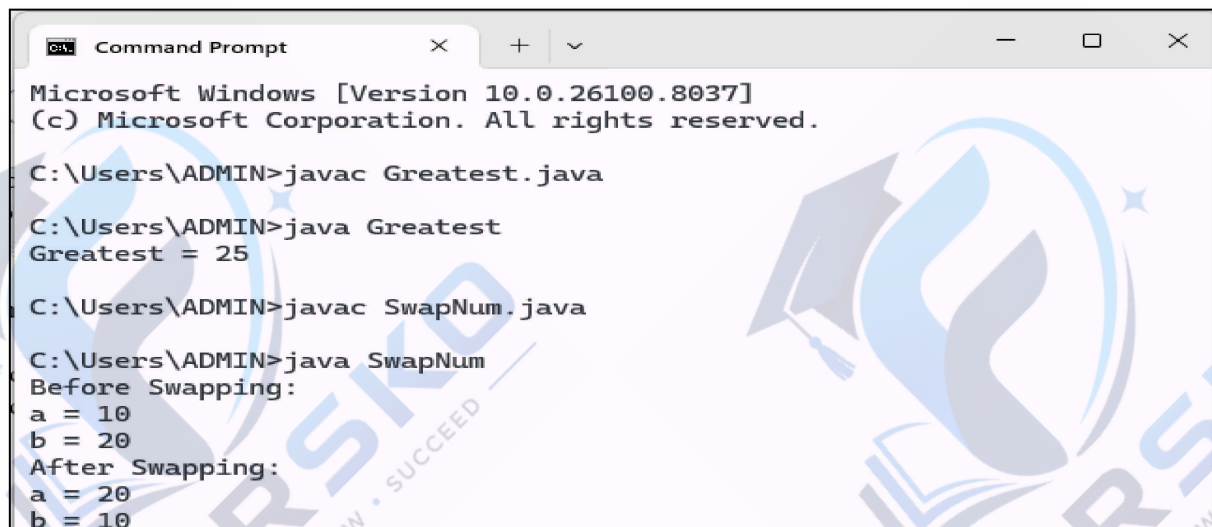
        System.out.println("Before Swapping:");
        System.out.println("a = " + a);
        System.out.println("b = " + b);

        a = a + b;
        b = a - b;
        a = a - b;

        System.out.println("After Swapping:");

        System.out.println("a = " + a);
        System.out.println("b = " + b);
    }
}
```

Output:



```
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ADMIN>javac Greatest.java

C:\Users\ADMIN>java Greatest
Greatest = 25

C:\Users\ADMIN>javac SwapNum.java

C:\Users\ADMIN>java SwapNum
Before Swapping:
a = 10
b = 20
After Swapping:
a = 20
b = 10
```

Practical No-02

Aim: Write a program in Java to determine whether an input number is an even number.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

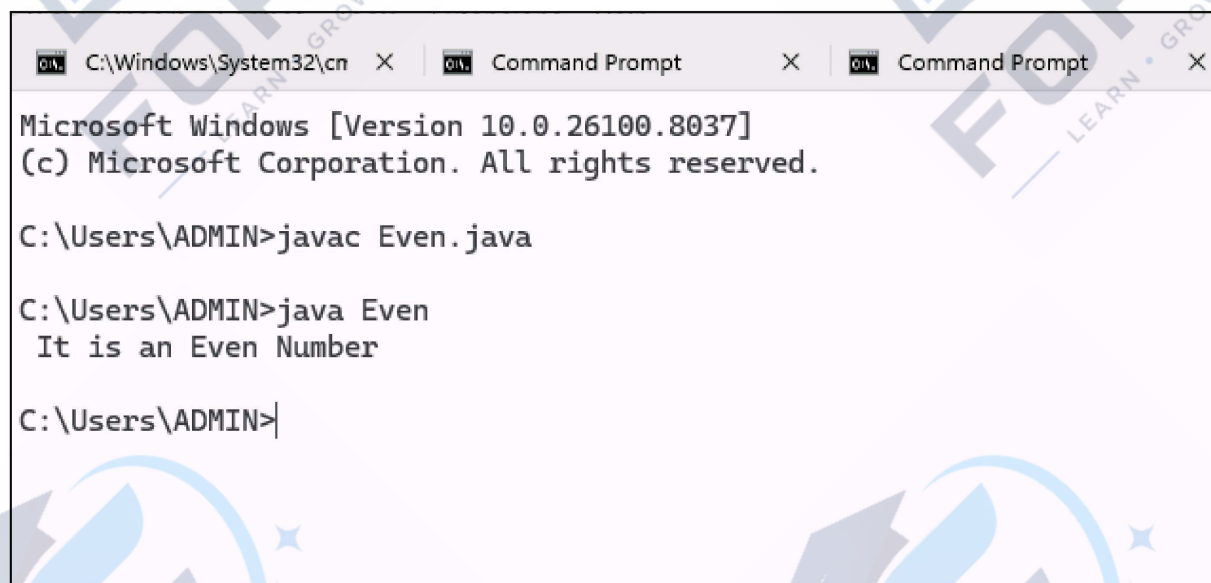
Id No: 71607

Date: 14/07/2026

Source Code:

```
public class Even {  
    public static void main(String[] args) {  
        int n = 6;  
  
        if (n % 2 == 0)  
            System.out.println(" It is an Even Number");  
        else  
            System.out.println(" It is an Odd Number");  
    }  
}
```

Output:



```
Microsoft Windows [Version 10.0.26100.8037]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\ADMIN>javac Even.java  
  
C:\Users\ADMIN>java Even  
It is an Even Number  
  
C:\Users\ADMIN>
```

Practical No-03

Aim: Write a program in Java to Find the Greatest of Three Numbers.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 14/07/2026

Source Code:

```
import java.util.Scanner;
```

```
public class Greatest {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        // Input three numbers  
        System.out.print("Enter first number: ");  
        int num1 = sc.nextInt();  
  
        System.out.print("Enter second number: ");  
        int num2 = sc.nextInt();  
  
        System.out.print("Enter third number: ");  
        int num3 = sc.nextInt();  
  
        // Logic to find greatest  
        int greatest;  
  
        if (num1 >= num2 && num1 >= num3) {  
            greatest = num1;  
        } else if (num2 >= num1 && num2 >= num3) {  
            greatest = num2;  
        } else {  
            greatest = num3;  
        }  
  
        // Output result  
        System.out.println("The greatest number is: " + greatest);  
  
        sc.close();  
    }  
}
```

Output:

```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>javac Greatest.java

C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>java Greatest
Enter first number: 14
Enter second number: 15
Enter third number: 23
The greatest number is: 23

C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>|
```

Practical No-04

Aim: Write a program in Java to Switch Case Statement

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 14/07/2026

Source Code:

```
import java.util.Scanner;

public class UpcommingMovies
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter Number from 1 to 4:");
        int c = sc.nextInt();

        switch(c)
        {
            case 1:

                System.out.println("Spider Man: Brand New Day\nRelease Date: 31 July 2026");

                break;

            case 2:

                System.out.println("Avengers : End Game\nRelease Date: 28 September 2026");

                break;

            case 3:

                System.out.println("Avengers : Dooms Day\nRelease Date: 18 December 2026");

                break;

            case 4:

                System.out.println("Avengers : BattleGround\n Release Date: Late 2028");

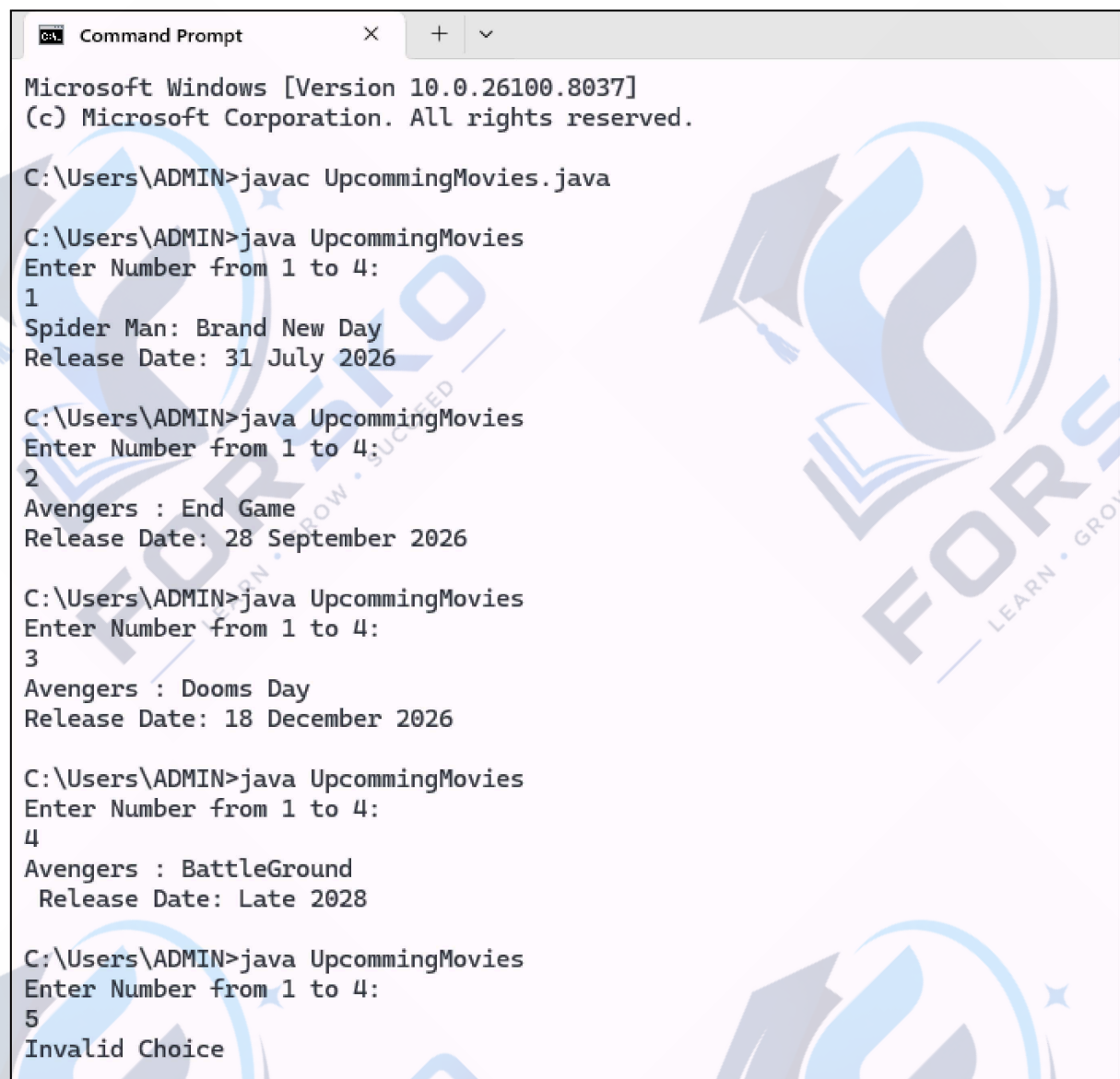
                break;
```

default :

```
System.out.println("Invalid Choice");
```

```
}  
  
}  
  
}
```

Output:



```
Microsoft Windows [Version 10.0.26100.8037]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\ADMIN>javac UpcommingMovies.java  
  
C:\Users\ADMIN>java UpcommingMovies  
Enter Number from 1 to 4:  
1  
Spider Man: Brand New Day  
Release Date: 31 July 2026  
  
C:\Users\ADMIN>java UpcommingMovies  
Enter Number from 1 to 4:  
2  
Avengers : End Game  
Release Date: 28 September 2026  
  
C:\Users\ADMIN>java UpcommingMovies  
Enter Number from 1 to 4:  
3  
Avengers : Dooks Day  
Release Date: 18 December 2026  
  
C:\Users\ADMIN>java UpcommingMovies  
Enter Number from 1 to 4:  
4  
Avengers : BattleGround  
Release Date: Late 2028  
  
C:\Users\ADMIN>java UpcommingMovies  
Enter Number from 1 to 4:  
5  
Invalid Choice
```

Practical No-05

Aim: Write a program in Java to Print Factorial of number

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 14/07/2026

Source Code:

```
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

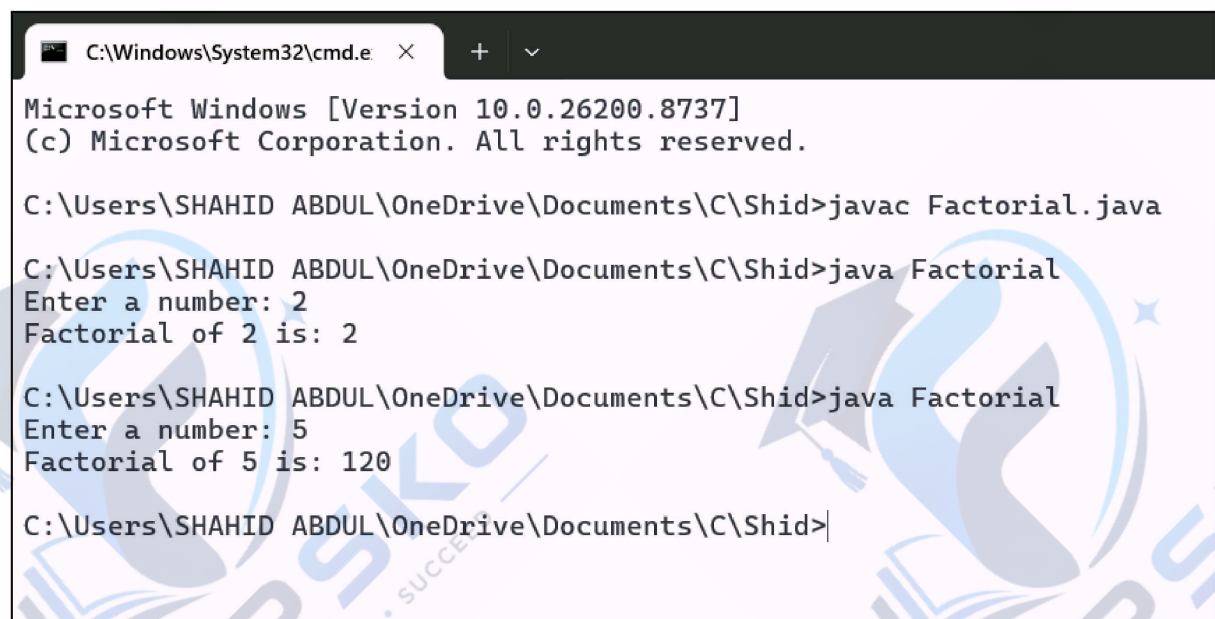
        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        long fact = 1;

        for (int i = 1; i <= num; i++) {
            fact *= i;
        }
        System.out.println("Factorial of " + num + " is: " + fact);

        sc.close();
    }
}
```

Output:



```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>javac Factorial.java

C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>java Factorial
Enter a number: 2
Factorial of 2 is: 2

C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>java Factorial
Enter a number: 5
Factorial of 5 is: 120

C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>
```

Practical No-06

Aim: Write a program in Java to Print Prime Number

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

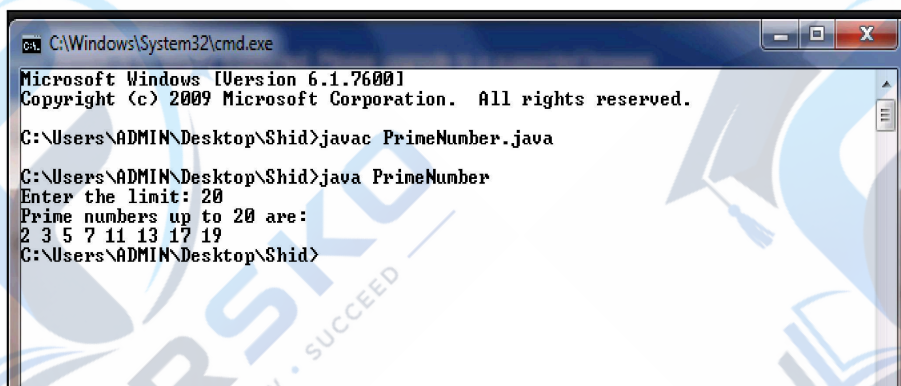
Date: 14/07/2026

Source Code:

```
import java.util.Scanner;
```

```
public class PrimeNumber {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter the limit: ");  
        int limit = sc.nextInt();  
        System.out.println("Prime numbers up to " + limit + " are:");  
  
        for (int i = 2; i <= limit; i++) {  
            boolean isPrime = true;  
  
            for (int j = 2; j <= Math.sqrt(i); j++) {  
                if (i % j == 0) {  
                    isPrime = false;  
                    break;  
                }  
            }  
            if (isPrime) {  
                System.out.print(i + " ");  
            }  
        }  
        sc.close();  
    }  
}
```

Output:



```
C:\Windows\System32\cmd.exe  
Microsoft Windows [Version 6.1.7600]  
Copyright (c) 2009 Microsoft Corporation. All rights reserved.  
  
C:\Users\ADMIN\Desktop\Shid>javac PrimeNumber.java  
C:\Users\ADMIN\Desktop\Shid>java PrimeNumber  
Enter the limit: 20  
Prime numbers up to 20 are:  
2 3 5 7 11 13 17 19  
C:\Users\ADMIN\Desktop\Shid>
```

Practical No-07

Aim: Write a program in Java to Generate Fibonacci Series

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 14/07/2026

Source Code:

```
import java.util.Scanner;

public class FS {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of terms: ");
        int n = sc.nextInt();

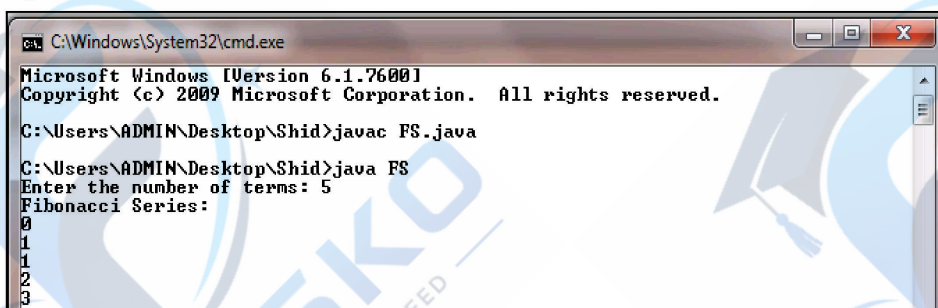
        int first = 0, second = 1;

        System.out.println("Fibonacci Series:");

        for (int i = 1; i <= n; i++) {
            System.out.print(first + " ");

            int next = first + second;
            first = second;
            second = next;
        }
        sc.close();
    }
}
```

Output:

A screenshot of a Windows command prompt window. The title bar reads "C:\Windows\System32\cmd.exe". The window content shows the following text:
Microsoft Windows [Version 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.
C:\Users\ADMIN\Desktop\Shid>javac FS.java
C:\Users\ADMIN\Desktop\Shid>java FS
Enter the number of terms: 5
Fibonacci Series:
0
1
1
2
3
The output shows the first five terms of the Fibonacci sequence: 0, 1, 1, 2, 3.

Practical No-08

Aim: Write a Java Program for Printing Multiplication table using do-while statement.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 14/07/2026

Source Code:

```
import java.util.Scanner;

public class Mul
{
    public static void main(String args [] )
    {
        Scanner sc = new Scanner(System.in);
        int num,i;

        System.out.print("Enter Number : ");
        num = sc.nextInt();
        i = 1;

        do
        {
            System.out.println(num + "x" +i+ "=" + (num * i));

            i ++;

        }
        while (i <= 10);

        sc.close ();
    }
}
```

Output:



```
C:\Windows\System32\cmd.exe

C:\Users\ADMIN\Desktop\Shid>java Mul
Enter Number :
2
2x1=2
2x2=4
2x3=6
2x4=8
2x5=10
2x6=12
2x7=14
2x8=16
2x9=18
2x10=20

C:\Users\ADMIN\Desktop\Shid>
```

Practical No-09

Aim: Write a Program in java by using a while loop to reverse the inputted number.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 29/07/2026

Source Code:

```
import java.util.Scanner;

public class ReverseNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

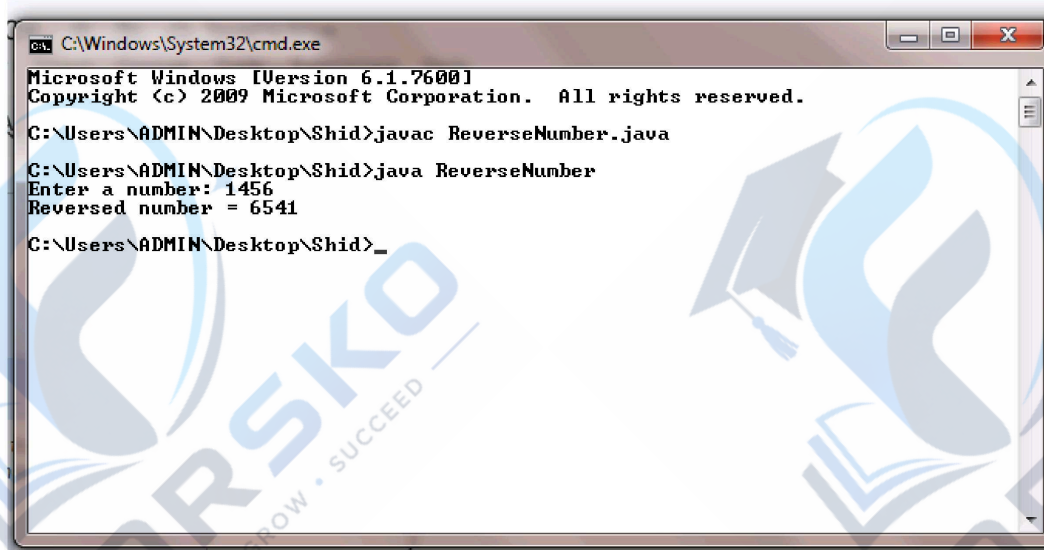
        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int reverse = 0;

        while (num != 0) {
            int digit = num % 10;
            reverse = reverse * 10 + digit;
            num = num / 10;
        }
        System.out.println("Reversed number = " + reverse);

        sc.close();
    }
}
```

Output:

A screenshot of a Windows command prompt window titled "C:\Windows\System32\cmd.exe". The window shows the execution of a Java program. The text in the window is as follows:

```
Microsoft Windows [Version 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\ADMIN\Desktop\Shid>javac ReverseNumber.java
C:\Users\ADMIN\Desktop\Shid>java ReverseNumber
Enter a number: 1456
Reversed number = 6541
C:\Users\ADMIN\Desktop\Shid>_
```

Practical No-10

Aim: Write a program in Java to Print following pattern

```
*  
**  
***  
****
```

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

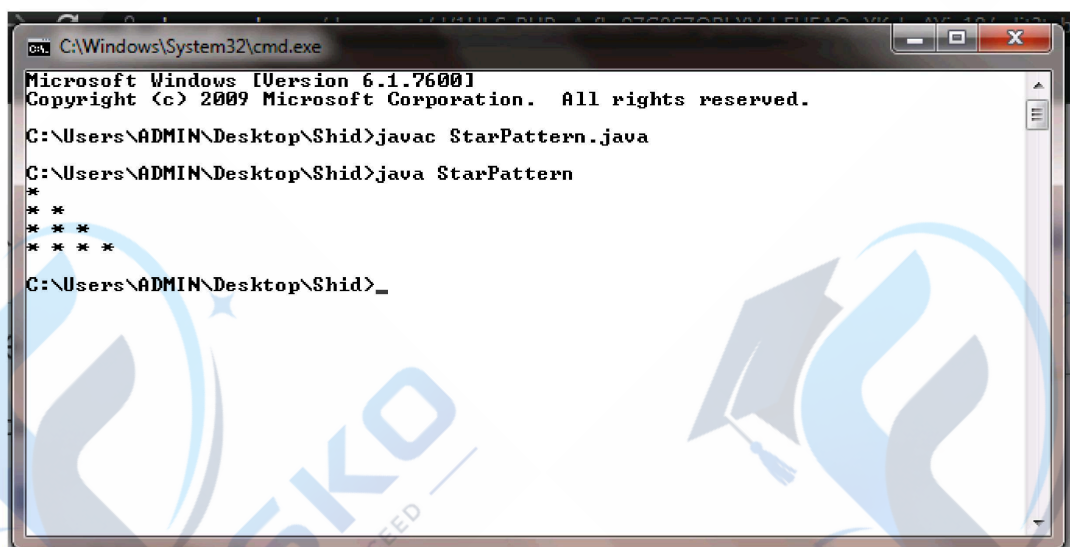
Id No: 71607

Date: 29/07/2026

Source Code:

```
public class StarPattern {  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 4; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print("* ");  
            }  
            System.out.println();  
        }  
    }  
}
```

Output:



```
C:\Windows\System32\cmd.exe  
Microsoft Windows [Version 6.1.7600]  
Copyright (c) 2009 Microsoft Corporation. All rights reserved.  
  
C:\Users\ADMIN\Desktop\Shid>javac StarPattern.java  
C:\Users\ADMIN\Desktop\Shid>java StarPattern  
*  
**  
***  
****  
C:\Users\ADMIN\Desktop\Shid>_
```

Practical No-11

Aim: Write a program in Java to print the following pattern:

1
12
123
1234
12345

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

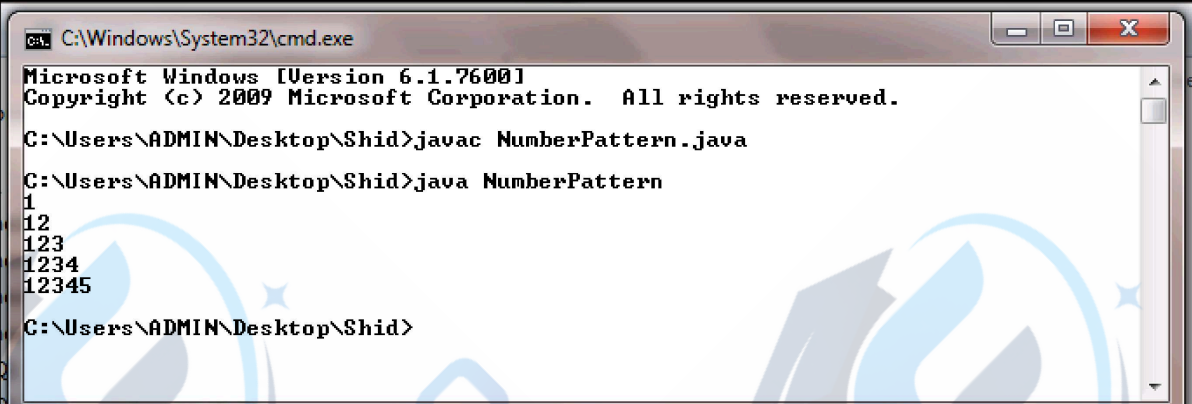
Id No: 71607

Date: 29/07/2026

Source Code:

```
public class NumberPattern {  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print(j);  
            }  
            System.out.println();  
        }  
    }  
}
```

Output:



```
C:\Windows\System32\cmd.exe  
Microsoft Windows [Version 6.1.7600]  
Copyright (c) 2009 Microsoft Corporation. All rights reserved.  
  
C:\Users\ADMIN\Desktop\Shid>javac NumberPattern.java  
C:\Users\ADMIN\Desktop\Shid>java NumberPattern  
1  
12  
123  
1234  
12345  
  
C:\Users\ADMIN\Desktop\Shid>
```

Practical No-12

Aim: Write a program in Java code to print the following pattern:

```
1
22
333
4444
55555
```

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

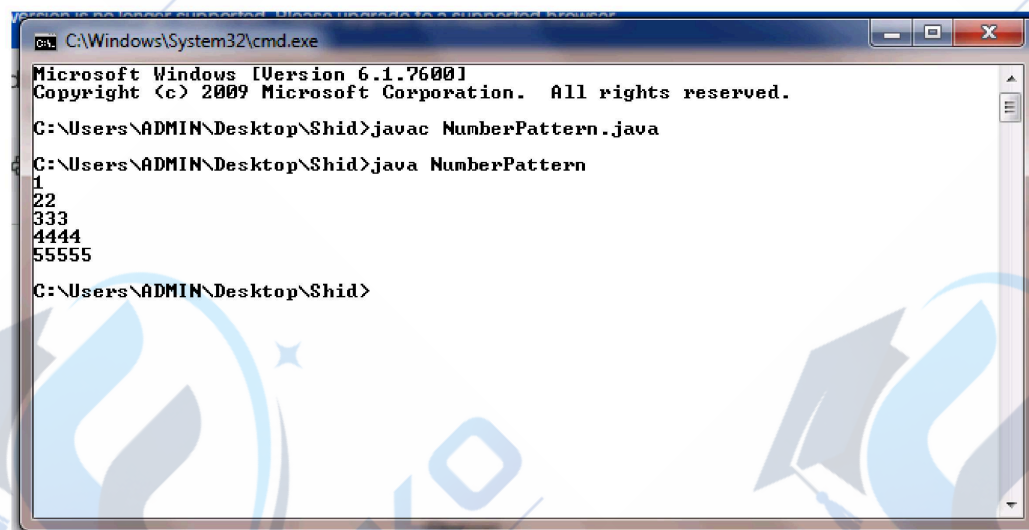
Date: 29/072026

Source Code :

```
public class NumberPattern {
    public static void main(String[] args) {

        for (int i = 1; i <= 5; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(i);
            }
            System.out.println();
        }
    }
}
```

Output :



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\ADMIN\Desktop\Shid>javac NumberPattern.java
C:\Users\ADMIN\Desktop\Shid>java NumberPattern
1
22
333
4444
55555
C:\Users\ADMIN\Desktop\Shid>
```

Practical No-13

Aim: Write a program in Java code to print the following pattern:

```
1
2 3
4 5 6
7 8 9 10
```

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 29/07/2026

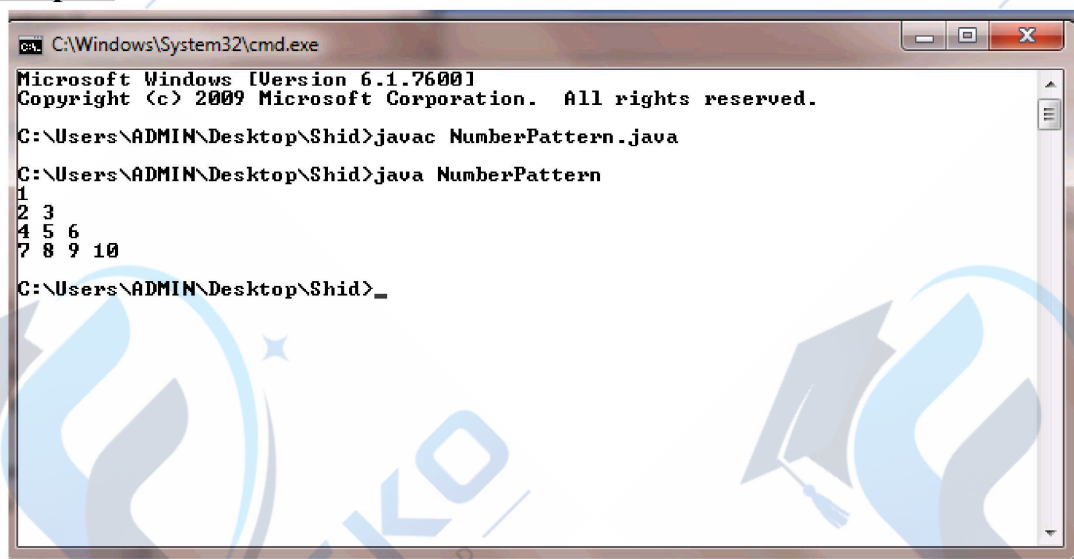
Source Code :

```
public class NumberPattern {
    public static void main(String[] args) {

        int num = 1;

        for (int i = 1; i <= 4; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(num + " ");
                num++;
            }
            System.out.println();
        }
    }
}
```

Output:



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\ADMIN\Desktop\Shid>javac NumberPattern.java
C:\Users\ADMIN\Desktop\Shid>java NumberPattern
1
2 3
4 5 6
7 8 9 10
C:\Users\ADMIN\Desktop\Shid>_
```

Practical No-14

Aim: Write a program in Java to demonstrate classes and objects.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 29/07/2026

Source Code :

```
class Student {
    int rollNo;
    String name;

    void getData(int r, String n) {
        rollNo = r;
        name = n;
    }

    void display() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
    }

    public static void main(String[] args) {
        Student s1 = new Student(); // Object 1
        Student s2 = new Student(); // Object 2

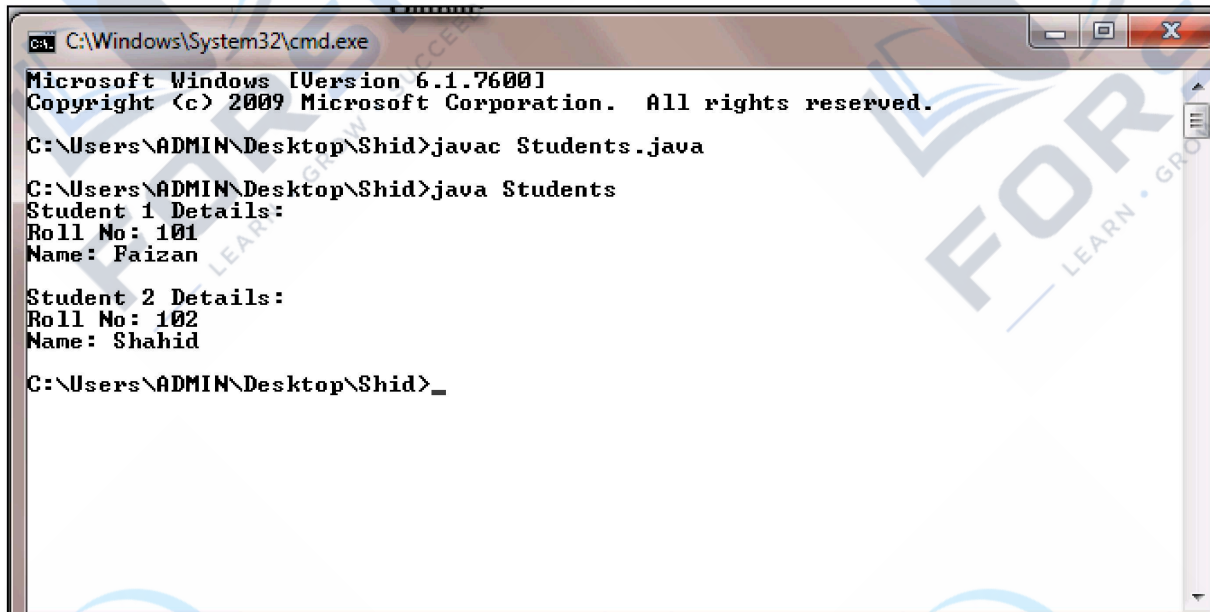
        s1.getData(101, "Faizan");
        s2.getData(102, "Shahid");

        System.out.println("Student 1 Details:");
        s1.display();

        System.out.println();

        System.out.println("Student 2 Details:");
        s2.display();
    }
}
```

Output:



A screenshot of a Windows Command Prompt window. The title bar reads 'C:\Windows\System32\cmd.exe'. The window content shows the following text:

```
Microsoft Windows [Version 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\ADMIN\Desktop\Shid>javac Students.java

C:\Users\ADMIN\Desktop\Shid>java Students
Student 1 Details:
Roll No: 101
Name: Faizan

Student 2 Details:
Roll No: 102
Name: Shahid

C:\Users\ADMIN\Desktop\Shid>_
```

Practical No-15

Aim: Write a program in Java to demonstrate method overloading.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

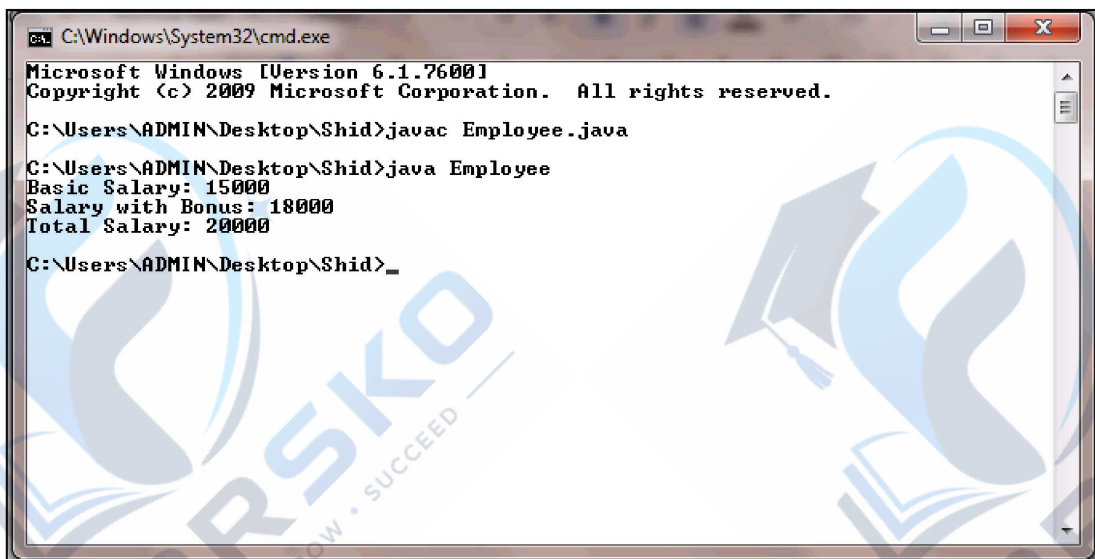
Id No: 71607

Date: 30/07/2026

Source Code :

```
class Employee {  
  
    void salary() {  
        System.out.println("Basic Salary: 15000");  
    }  
  
    void salary(int bonus) {  
        System.out.println("Salary with Bonus: " + (15000 + bonus));  
    }  
  
    void salary(int bonus, int allowance) {  
        System.out.println("Total Salary: " + (15000 + bonus + allowance));  
    }  
  
    public static void main(String[] args) {  
        Employee obj = new Employee();  
  
        obj.salary();  
        obj.salary(3000);  
        obj.salary(3000, 2000);  
    }  
}
```

Output:



```
cmd: C:\Windows\System32\cmd.exe  
Microsoft Windows [Version 6.1.7600]  
Copyright (c) 2009 Microsoft Corporation. All rights reserved.  
  
C:\Users\ADMIN\Desktop\Shid>javac Employee.java  
  
C:\Users\ADMIN\Desktop\Shid>java Employee  
Basic Salary: 15000  
Salary with Bonus: 18000  
Total Salary: 20000  
  
C:\Users\ADMIN\Desktop\Shid>
```

Practical No-16

Aim: Write a program in Java to demonstrate constructors.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 06/08/2026

Source Code :

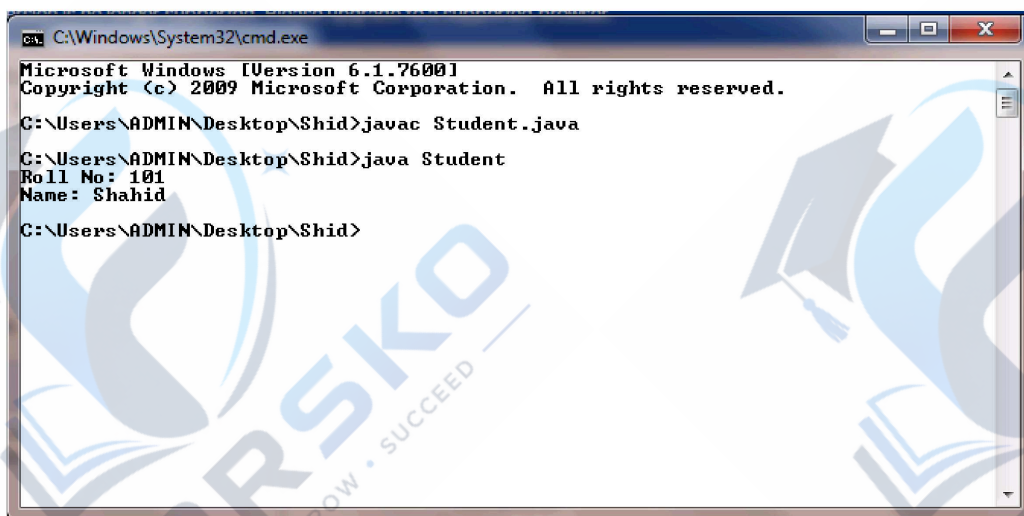
```
class Student
{
    int rollNo;
    String name;
    Student()
    {
        rollNo = 101;
        name = "Shahid";
    }

    void display()
    {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
    }

    public static void main(String args[])
    {
        Student s = new Student();

        s.display();
    }
}
```

Output:



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\ADMIN\Desktop\Shid>javac Student.java
C:\Users\ADMIN\Desktop\Shid>java Student
Roll No: 101
Name: Shahid
C:\Users\ADMIN\Desktop\Shid>
```

Practical No-17

Aim: Write a program in Java to demonstrate constructor overloading.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 06/08/2026

Source Code :

```
class Book
{
    int price;
    String name;

    Book()
    {
        price = 200;
        name = "Java";
    }

    Book(int p)
    {
        price = p;
        name = "Python";
    }

    Book(int p, String n)
    {
        price = p;
        name = n;
    }

    void display()
    {
        System.out.println("Book Name: " + name);
        System.out.println("Price: " + price);
        System.out.println();
    }

    public static void main(String args[])
    {
        Book b1 = new Book();
```

```
Book b2 = new Book(300);  
Book b3 = new Book(500, "C++");  
  
b1.display();  
b2.display();  
b3.display();  
}  
}
```

Output :



```
C:\Windows\System32\cmd.exe  
Microsoft Windows [Version 6.1.7600]  
Copyright (c) 2009 Microsoft Corporation. All rights reserved.  
  
C:\Users\ADMIN\Desktop\Shid>javac Book.java  
  
C:\Users\ADMIN\Desktop\Shid>java Book  
Book Name: Can we be Strangers Again ?  
Price: 200  
  
Book Name: Python  
Price: 300  
  
Book Name: C++  
Price: 500  
  
C:\Users\ADMIN\Desktop\Shid>_
```

Practical No-18

Aim: Write a program in Java to demonstrate single inheritance

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

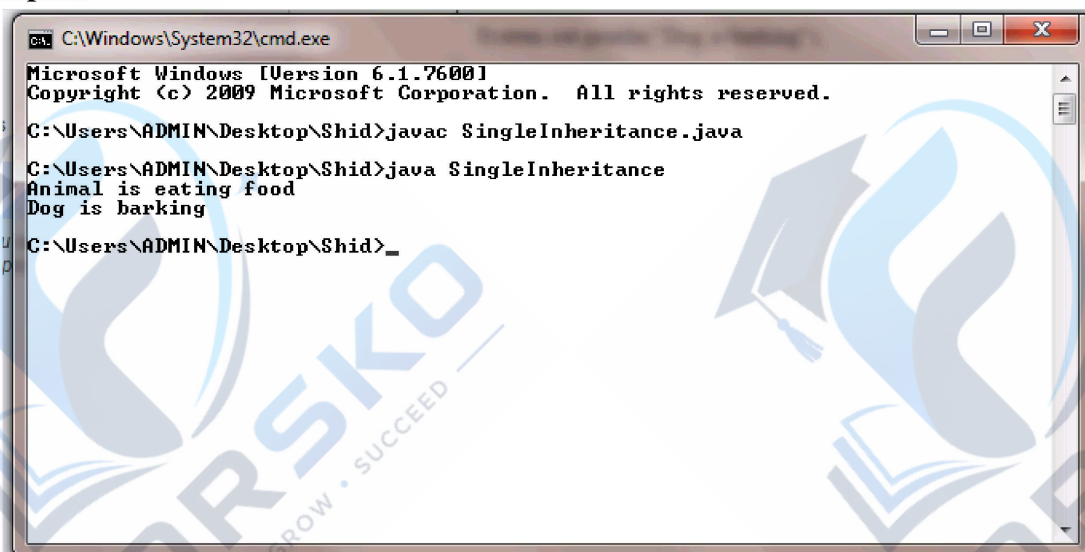
Date: 06/08/2026

Source Code :

```
class Animal
{
    void eat()
    {
        System.out.println("Animal is eating food");
    }
}
class Dog extends Animal
{
    void bark()
    {
        System.out.println("Dog is barking");
    }
}
public class SingleInheritance
{
    public static void main(String[] args)
    {
        Dog d = new Dog();

        d.eat();
        d.bark();
    }
}
```

Output:



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\ADMIN\Desktop\Shid>javac SingleInheritance.java
C:\Users\ADMIN\Desktop\Shid>java SingleInheritance
Animal is eating food
Dog is barking
C:\Users\ADMIN\Desktop\Shid>_
```

Practical No-19

Aim: Write a program in Java to demonstrate method overriding.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 06/08/2026

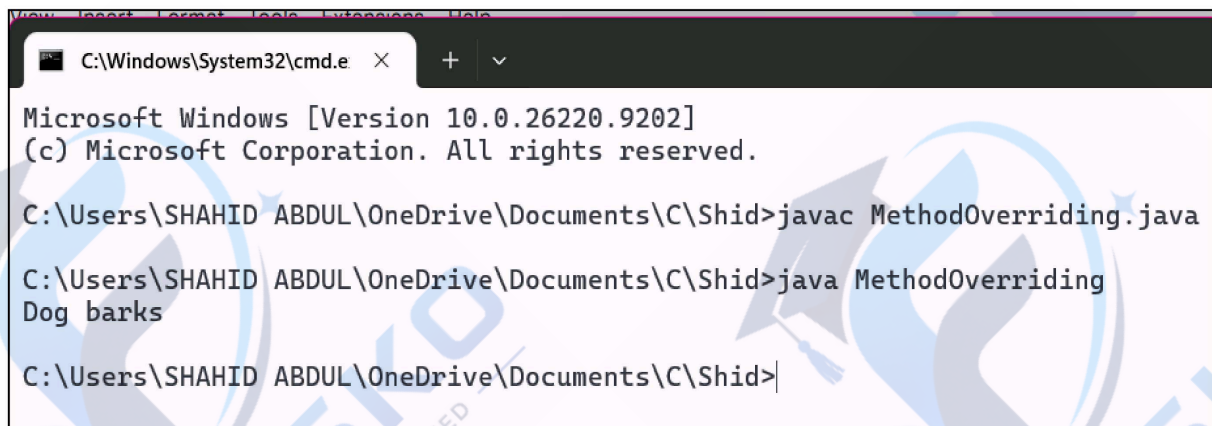
Source Code :

```
class Animal {
    void sound() {
        System.out.println("Animal makes a sound");
    }
}

class Dog extends Animal {
    // Overriding the sound() method
    @Override
    void sound() {
        System.out.println("Dog barks");
    }
}

public class MethodOverriding {
    public static void main(String[] args) {
        Animal obj = new Dog();
        obj.sound();
    }
}
```

Output:



```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26220.9202]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>javac MethodOverriding.java

C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>java MethodOverriding
Dog barks

C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>
```

Practical No-20

Aim: Write a program in Java to demonstrate Interface.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

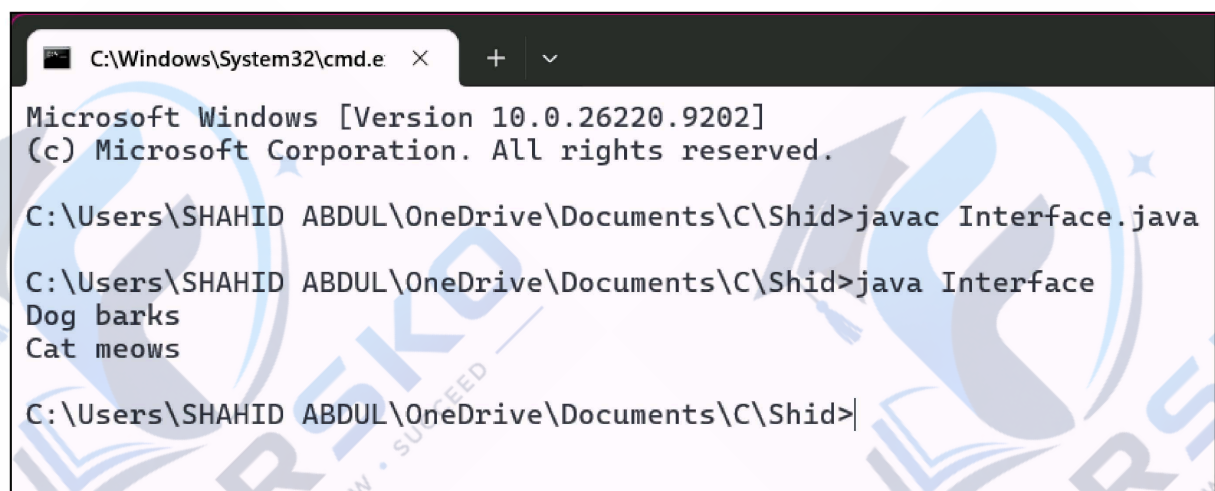
Id No: 71607

Date: 06/08/2026

Source Code :

```
interface Animal {  
    void sound();  
}  
  
class Dog implements Animal {  
    public void sound() {  
        System.out.println("Dog barks");  
    }  
}  
  
class Cat implements Animal {  
    public void sound() {  
        System.out.println("Cat meows");  
    }  
}  
  
public class Interface {  
    public static void main(String[] args) {  
        Animal a1 = new Dog();  
        Animal a2 = new Cat();  
  
        a1.sound();  
        a2.sound();  
    }  
}
```

Output:



```
C:\Windows\System32\cmd.e  X  +  v  
Microsoft Windows [Version 10.0.26220.9202]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>javac Interface.java  
  
C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>java Interface  
Dog barks  
Cat meows  
  
C:\Users\SHAHID ABDUL\OneDrive\Documents\C\Shid>|
```

Practical No-21

Aim: Write a Python program to create, append and remove lists.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 29/07/2026

Source Code :

```
# Create a list
```

```
Mobile_Apps = ["Instagram", "WhatsApp", "X", "Free Fire"]
```

```
print("Original List:", Mobile_Apps)
```

```
# Append an item
```

```
print("Original List:", Mobile_Apps)
```

```
Mobile_Apps.append("Telegram")
```

```
print("After Append:", Mobile_Apps)
```

```
# Remove an item
```

```
Mobile_Apps.remove("X")
```

```
print("After Remove:", Mobile_Apps)
```

Output :

```
*** Original List: ['Instagram', 'WhatsApp', 'X', 'Free Fire']
Original List: ['Instagram', 'WhatsApp', 'X', 'Free Fire']
After Append: ['Instagram', 'WhatsApp', 'X', 'Free Fire', 'Telegram']
After Remove: ['Instagram', 'WhatsApp', 'Free Fire', 'Telegram']
```

Practical No-22

Aim: Python program to check if a substring is present in a given string.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 30/07/2026

Source Code :

```
# Python program to check if a substring is present in a given string
```

```
string = input("Enter the main string: ")
```

```
substring = input("Enter the substring: ")
```

```
if substring in string:
```

```
    print("Substring is present in the string.")
```

```
else:
```

```
    print("Substring is not present in the string.")
```

Output:

```
... Enter the main string: Hello , World !  
Enter the substring: World  
Substring is present in the string.
```

```
... Enter the main string: Hello Sir Ji !  
Enter the substring: World  
Substring is not present in the string.
```

Practical No-23

Aim: Write a program to demonstrate working with tuples in Python.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 3/08/2026

Source Code :

```
# Program to demonstrate working with tuples in Python
```

```
# Creating a tuple
```

```
my_tuple = (10, 20, 30, 40, 50)
```

```
print("Tuple:", my_tuple)
```

```
# Accessing tuple elements
```

```
print("\nFirst element:", my_tuple[0])
```

```
print("\nThird element:", my_tuple[2])
```

```
# Slicing a tuple
```

```
print("\nFirst three elements:", my_tuple[0:3])
```

```
# Length of tuple
```

```
print("\nLength of tuple:", len(my_tuple))
```

```
# Checking an element
```

```
if 30 in my_tuple:
```

```
    print("\n30 is present in the tuple.")
```

```
# Counting an element
```

```
print("\nCount of 20:", my_tuple.count(20))
```

```
# Finding the index of an element
```

```
print("\nIndex of 40:", my_tuple.index(40))
```

```
# Concatenating tuples
```

```
tuple2 = (60, 70)
new_tuple = my_tuple + tuple2
print("\nAfter concatenation:", new_tuple)

# Repeating a tuple
print("\nRepeated tuple:", tuple2 * 2)
```

Output :

```
... Tuple: (10, 20, 30, 40, 50)

First element: 10

Third element: 30

First three elements: (10, 20, 30)

Length of tuple: 5

30 is present in the tuple.

Count of 20: 1

Index of 40: 3

After concatenation: (10, 20, 30, 40, 50, 60, 70)

Repeated tuple: (60, 70, 60, 70)
```

Practical No-24

Aim: Write a program to demonstrate working with dictionaries in Python.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 03/08/2026

Source Code :

```
# Program to demonstrate working with dictionaries in Python
```

```
Books = { "Preview No ": 101 , "Book Name": "Can we become Stranger Again ?" , "Price":  
200 , "Tag": "Self Story" }
```

```
# Display the Dictionary
```

```
print("\nBooks Dictionary:", Books)
```

```
# Accessing the Values
```

```
print("\nBook Name:", Books["Book Name"])
```

```
#Updating Value
```

```
Books["Price"] = 350
```

```
#Displaying the Updated Dictionary
```

```
print("\nUpdated Dictionaray:", Books)
```

```
#Removing Value
```

```
Books.pop("Tag")
```

```
#Reading Through the Dictionary
```

```
print("\nDictionary Elements")
```

```
for key,value in Books.items():
```

```
print(key,":", value)
```

Output :

```
'''
```

```
Books Dictionary: {'Preview No ': 101, 'Book Name': 'Can we become Stranger Again ?', 'Price': 200, 'Tag': 'Self Story'}
```

```
Book Name: Can we become Stranger Again ?
```

```
Updated Dictionaray: {'Preview No ': 101, 'Book Name': 'Can we become Stranger Again ?', 'Price': 350, 'Tag': 'Self Story'}
```

```
Dictionary Elements
```

```
Preview No : 101
```

```
Book Name : Can we become Stranger Again ?
```

```
Price : 350
```

Practical No-25

Aim: Write a Python program to find the largest of three numbers.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 03/08/2026

Source Code :

```
# Program to find the largest of three numbers
```

```
a = int(input("\nEnter first number: "))
```

```
b = int(input("\nEnter second number: "))
```

```
c = int(input("\nEnter third number: "))
```

```
if a >= b and a >= c:
```

```
    largest = a
```

```
elif b >= a and b >= c:
```

```
    largest = b
```

```
else:
```

```
    largest = c
```

```
print("\nLargest number is:", largest)
```

Output:

```
***
```

```
Enter first number: 15
```

```
Enter second number: 16
```

```
Enter third number: 18
```

```
Largest number is: 18
```

Practical No-26

Aim: Write a Python program to construct the following pattern using nested for loop.

```
*  
**  
***  
****
```

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 05/08/2026

Source Code :

```
# Program to print the following pattern using nested for loop  
# *  
# **  
# ***  
# ****  
  
for i in range(1, 5):  
    for j in range(1, i + 1):  
        print("*", end="")  
    print()
```

Output:

```
*** *  
** **  
* ** *  
* ** **
```

Practical No-27

Aim: Write a Python program to print prime numbers less than 20.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 05/08/2026

Source Code :

```
# Program to print prime numbers less than 20
```

```
print("Prime numbers less than 20\n")
```

```
for num in range(2, 20):
```

```
    count = 0
```

```
    for i in range(1, num + 1):
```

```
        if num % i == 0:
```

```
            count += 1
```

```
    if count == 2:
```

```
        print(num, end="\n")
```

Output :

```
*** Prime numbers less than 20
2
3
5
7
11
13
17
19
```

Practical No-28

Aim: Write a Python program to find factorial of a number using recursion.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 05/08/2026

Source Code :

```
# Program to find factorial of a number using recursion
```

```
def factorial(n):  
    if n == 0 or n == 1:  
        return 1  
    else:  
        return n * factorial(n - 1)  
  
num = int(input("Enter a number: "))  
  
result = factorial(num)  
  
print("Factorial of", num, "is:", result)
```

Output :

```
... Enter a number: 5  
Factorial of 5 is: 120
```

Practical No-29

Aim: Write a Python program to map two lists into a dictionary.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 07/08/2026

Source Code :

```
# Program to map two lists into a dictionary
```

```
names = ["Amit", "Rahul", "Priya", "Neha"]
```

```
marks = [85, 90, 78, 88]
```

```
student = dict(zip(names, marks))
```

```
print("Student Marks:")
```

```
print(student)
```

Output:

```
*** Student Marks:
{'Amit': 85, 'Rahul': 90, 'Priya': 78, 'Neha': 88}
```

Practical No-30

Aim: Write a Python program to count the frequency of words appearing in a string using a dictionary.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 08/08/2026

Source Code :

```
# Program to count the frequency of words in a string
```

```
string = input("Enter a string: ")
```

```
words = string.split()
```

```
frequency = {}
```

```
for word in words:
```

```
    if word in frequency:
```

```
        frequency[word] += 1
```

```
    else:
```

```
        frequency[word] = 1
```

```
print("Word Frequency:")
```

```
for word, count in frequency.items():
```

```
    print(word, ":", count)
```

Output:

```
Enter a string: python is easy python is powerful
Word Frequency:
python : 2
is : 2
easy : 1
powerful : 1
```

Practical No-31

Aim: Write a Python program to create a dictionary with key as first character and value as words starting with that character.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 08/08/2026

Source Code :

```
# Program to create a dictionary with
# first character as key and words as values

words = ["apple", "ant", "ball", "bat", "cat", "car", "dog"]

dictionary = {}

for word in words:
    first_char = word[0]

    if first_char in dictionary:
        dictionary[first_char].append(word)
    else:
        dictionary[first_char] = [word]

print("Dictionary:")
print(dictionary)
```

Output:

```
... Dictionary:
{'a': ['apple', 'ant'], 'b': ['ball', 'bat'], 'c': ['cat', 'car'], 'd': ['dog']}
```

Practical No-32

Aim: Write a Python program to read the contents of a file in reverse order.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 08/08/2026

Source Code :

```
# Create a sample file

with open("sample.txt", "w") as file:
    file.write("Hello Python\nWelcome to Programming")

# Program to read the contents of a file in reverse order

file = open("sample.txt", "r")

content = file.read()

reverse_content = content[::-1]

print("File contents in reverse order:")
print(reverse_content)

file.close()
```

Output :

```
... File contents in reverse order:
   gnimmargorP ot emocleW
   nohtyP olleH
```

Practical No-33

Aim: Write a Python program to map two lists into a dictionary.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 10/08/2026

Source Code :

```
# Program to map two lists into a dictionary
```

```
MyBooks = ["Can we be Stranger Again ?", "We are in Love", "The Art of Forgiving",  
"BestFriend to Strangers"]
```

```
BookPrice = [300, 300, 350, 400]
```

```
result = dict(zip(MyBooks, BookPrice))
```

```
print("Mapping Dictionary")
```

```
for book, price in result.items():  
    print(book, ":", price)
```

Output:

```
... Mapping Dictionary  
Can we be Stranger Again ? : 300  
We are in Love : 300  
The Art of Forgiving : 350  
BestFriend to Strangers : 400
```

Practical No-34

Aim: Write a Python Program to demonstrate the class and object.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 11/08/2026

Source Code :

```
# Define a class
class Youtubers:
    def show(self):
        print("YouTuber Name: Not Your Type")
        print("Channel: Not Your Type")
        print("Subscribers: 10 Million")
        print("Content: Animated, Relatable and Comedy Videos")

# Create an object
youtuber1 = Youtubers()

# Call the method
youtuber1.show()
```

Output:

```
... YouTuber Name: Not Your Type
    Channel: Not Your Type
    Subscribers: 10 Million
    Content: Animated, Relatable and Comedy Videos
```

Practical No-35

Aim: Write a Python Program to demonstrate constructor methods.

Name: Abdul Shahid

Class: B.Sc -III (SEM-V)

Subject: Computer Science

Id No: 71607

Date: 11/08/2026

Source Code :

```
class Youtuber:
    def __init__(self, name, subscribers):
        self.name = name
        self.subscribers = subscribers

    def display(self):
        print("Youtuber Name:", self.name)
        print("\nSubscribers:", self.subscribers)

# Creating object
y1 = Youtuber("Not Your Type", "10 Million")

# Display details
y1.display()
```

Output :

```
*** Youtuber Name: Not Your Type
    Subscribers: 10 Million
```